

RaahSetu

AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)

Ministry / Organization	Ministry of Development of North Eastern Region (MDoNER)
Problem Statement Number	SIH26002
Problem Statement Title	AI-Based Smart Logistics and Accessibility Intelligence Platform for NER
Idea Title	RaahSetu — Explainable Risk-Aware Logistics Routing
Team Name	[Team name as registered on portal]
PS Category	Software



Idea / Proposed Solution

The problem in one line

NH-6 alone is blocked 15–20 times each monsoon, stranding heavy trucks for 3–5 days; standard navigation apps still optimise only for speed, blind to landslides, floods and single-access corridors.

Our solution

A risk-aware routing engine and operations workspace that plans heavy-vehicle journeys around verified road hazards — and shows the evidence behind every decision.

Two routes, one decision

Every journey compares the fastest route with a risk-aware alternative: time cost, distance and hazard exposure side by side.

Evidence before closure

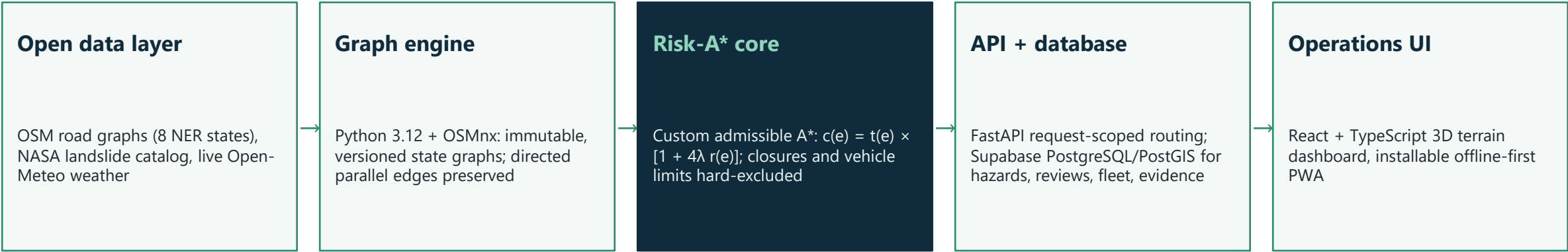
Field reports carry corroboration counts; only a reviewer-verified closure changes routing. A single false report can never block a road.

Works where connectivity fails

Offline report queue with automatic sync; missing evidence stays “unknown” — never silently treated as safe.

Differentiator: risk enters the cost function itself — $c(e) = t(e) \times [1 + 4\lambda \cdot r(e)]$ — so the tradeoff between speed and safety is quantified, explainable and auditable.

Technical Architecture



Middle box (Risk-A core) highlighted: the algorithm is the product — everything else serves it.*

Technology stack

Frontend	React, TypeScript, React Three Fiber (3D terrain), PWA service worker
Backend	Python 3.12, FastAPI, Pydantic validation
Spatial engine	OSMnx / NetworkX graph processing, PostGIS spatial indexing
Database	Supabase PostgreSQL + PostGIS (row-level security)
Deployment	Docker, Kubernetes on VMs, versioned graph storage
External data	OpenStreetMap (ODbL), NASA landslide catalog, Open-Meteo

Verified working: 45 automated tests, 300 randomized A* vs Dijkstra oracle checks, 14-step live cloud end-to-end workflow, 8-state OSM data acquisition (273,494 road ways).

Use Cases

Primary — Emergency relief fleets

NDRF / SDRF and health supply chains moving relief material during extreme weather: pre-departure risk routing around landslide- and flood-affected corridors, with detour cost known up-front.

Secondary — Commercial heavy logistics

Truck operators on NER corridors (NH-6, NH-2, NH-10): safer route selection that respects vehicle limits and surfaces, cutting fuel and delay penalties in a region where logistics runs 20–30% costlier.

Tertiary — State transport authorities

A verifiable incident pipeline: corroborated field reports → reviewer approval → versioned closure events → district connectivity status, replacing unverified forwards with auditable evidence.

Special mode — Single-access risk

When a sole access route to a cut-off district is verified as blocked, the platform flags connectivity status and alerts authorities for transshipment or air-drop coordination (rail / river-terminal alternatives mapped next).

Dependencies and Show Stoppers

Dependencies

- Official coordinate-level blackspot / eDAR and MoRTH accident data (verified road matching).
- IMD predictive weather feeds for pre-departure hazard probability (live Open-Meteo today).
- Transport-domain expert review of provisional hazard-to-road matches.
- Cloud VM / Kubernetes access for public HTTPS deployment.

Show stoppers — and how the design answers them

Live traffic and hazard APIs are paid

Open OSM graphs + rule-based risk scoring + keyless Open-Meteo weather: zero paid API dependency in the prototype.

Unverified crowd reports can poison routing

Credibility layer: corroboration counted within 1 km / 6 h; only reviewer-approved closures change routing — a lone false report is structurally powerless.

NER road network too large for the browser

State-wise versioned graph loading; the browser only receives the active state's bounded road set.

Connectivity drops in remote corridors

Offline-first PWA: reports queue locally with photo evidence and sync automatically when the network returns.